

Probability Practice Problems

Problem 1 (Joint and Marginal Probabilities)

A weather monitoring system tracks Temperature (T: hot/cold) and Wind (W: strong/weak). The joint probability distribution is:

T	W	P(T,W)
hot	strong	0.15
hot	weak	0.35
cold	strong	0.20
cold	weak	0.30

Calculate: a) $P(\text{hot})$, b) $P(\text{strong})$, c) $P(\text{hot OR strong})$

Problem 2 (Conditional Probability)

Using the table from Problem 1, calculate: a) $P(\text{hot} \mid \text{strong})$ b) $P(\text{strong} \mid \text{cold})$ c) $P(W \mid \text{hot})$ - provide the full conditional distribution

Problem 3 (Bayes' Rule)

A medical test for disease D has the following properties:

- $P(D) = 0.01$ (1% of population has the disease)
- $P(+\text{test} \mid D) = 0.95$ (sensitivity)
- $P(-\text{test} \mid \neg D) = 0.90$ (specificity)

If a patient tests positive, what is $P(D \mid +\text{test})$?

Problem 4 (Independence)

Consider three events with the following probabilities:

- $P(A) = 0.4$, $P(B) = 0.5$
- $P(A, B) = 0.2$
- $P(C \mid A) = 0.3$, $P(C \mid \neg A) = 0.3$

a) Are A and B independent? Show your work. b) Are A and C independent? Explain.

Problem 5 (Three Variable Joint Distribution)

A smart home tracks: Occupancy (O), Temperature Setting (T), and Energy Usage (E), all binary.

O	T	E	P(O,T,E)
+o	high	high	0.24
+o	high	low	0.06
+o	low	high	0.08
+o	low	low	0.12
-o	high	high	0.10
-o	high	low	0.10
-o	low	high	0.05
-o	low	low	0.25

Calculate: a) $P(+o)$, b) $P(\text{high energy} \mid +o)$, c) $P(\text{high temp} \mid -o, \text{low energy})$

Problem 6 (Conditional Independence)

Given: $P(\text{Rain}) = 0.2$, $P(\text{Sprinkler} \mid \text{Rain}) = 0.01$, $P(\text{Sprinkler} \mid \neg \text{Rain}) = 0.4$, $P(\text{Wet} \mid \text{Rain}, \text{Sprinkler}) = 0.99$, $P(\text{Wet} \mid \text{Rain}, \neg \text{Sprinkler}) = 0.80$, $P(\text{Wet} \mid \neg \text{Rain}, \text{Sprinkler}) = 0.90$, $P(\text{Wet} \mid \neg \text{Rain}, \neg \text{Sprinkler}) = 0.01$

Calculate $P(\text{Wet})$ using marginalization.

Problem 7 (Bayes Net Structure)

A Bayes net has nodes A, B, C, D with edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$. Given: $P(A) = 0.3$, $P(B \mid A) = 0.8$, $P(B \mid \neg A) = 0.2$, $P(C \mid A) = 0.7$, $P(C \mid \neg A) = 0.4$, $P(D \mid B, C) = 0.9$, $P(D \mid B, \neg C) = 0.6$, $P(D \mid \neg B, C) = 0.5$, $P(D \mid \neg B, \neg C) = 0.1$

Calculate $P(D)$ by marginalizing over all parent variables.

Problem 8 (Inference by Enumeration)

Using the weather data:

Cloudy	Rain	P(C,R)
yes	yes	0.32
yes	no	0.08
no	yes	0.18
no	no	0.42

Calculate $P(\text{Cloudy} \mid \text{Rain})$ using: a) definition of conditional probability, b) normalization trick

Problem 9 (Chain Rule)

Given $P(A) = 0.4$, $P(B \mid A) = 0.7$, $P(B \mid \neg A) = 0.3$, $P(C \mid A, B) = 0.8$, $P(C \mid A, \neg B) = 0.5$, $P(C \mid \neg A, B) = 0.6$, $P(C \mid \neg A, \neg B) = 0.2$

Use the chain rule to calculate $P(A, B, C)$.

Problem 10 (Explaining Away)

A car won't start. Possible causes: Dead Battery (B) with $P(B) = 0.3$, or No Gas (G) with $P(G) = 0.1$. The car won't start if either is true: $P(\text{Won't Start} \mid B, G) = 1$, $P(\text{Won't Start} \mid B, \neg G) = 1$, $P(\text{Won't Start} \mid \neg B, G) = 1$, $P(\text{Won't Start} \mid \neg B, \neg G) = 0.05$

a) Calculate $P(\text{Won't Start})$ b) Given the car won't start, what is $P(B \mid \text{Won't Start})$? c) Given the car won't start AND we know there's gas, what is $P(B \mid \text{Won't Start}, G)$?

ANSWER KEY

Problem 1

- a) $P(\text{hot}) = P(\text{hot, strong}) + P(\text{hot, weak}) = 0.15 + 0.35 = \mathbf{0.50}$
b) $P(\text{strong}) = P(\text{hot, strong}) + P(\text{cold, strong}) = 0.15 + 0.20 = \mathbf{0.35}$
c) $P(\text{hot OR strong}) = P(\text{hot}) + P(\text{strong}) - P(\text{hot, strong}) = 0.50 + 0.35 - 0.15 = \mathbf{0.70}$

Problem 2

- a) $P(\text{hot} \text{ --- strong}) = P(\text{hot, strong}) / P(\text{strong}) = 0.15 / 0.35 = \mathbf{3/7 \approx 0.429}$
b) $P(\text{strong} \text{ --- cold}) = P(\text{cold, strong}) / P(\text{cold}) = 0.20 / 0.50 = \mathbf{0.40}$
c) $P(W \text{ --- hot})$:
- $P(\text{strong} \text{ --- hot}) = 0.15 / 0.50 = \mathbf{0.30}$
 - $P(\text{weak} \text{ --- hot}) = 0.35 / 0.50 = \mathbf{0.70}$

Problem 3

First find $P(+test)$:

$$P(+test) = P(+test|D) \cdot P(D) + P(+test|\neg D) \cdot P(\neg D) \quad (1)$$

$$= 0.95 \times 0.01 + 0.10 \times 0.99 \quad (2)$$

$$= 0.0095 + 0.099 = 0.1085 \quad (3)$$

Apply Bayes' rule:

$$P(D|+test) = \frac{P(+test|D) \cdot P(D)}{P(+test)} \quad (4)$$

$$= \frac{0.95 \times 0.01}{0.1085} = \frac{0.0095}{0.1085} = \mathbf{0.0876} \quad (5)$$

Problem 4

- a) For independence, need $P(A,B) = P(A) \times P(B)$
- $P(A) \times P(B) = 0.4 \times 0.5 = 0.2$
 - $P(A,B) = 0.2$
 - Since they're equal, A and B **ARE independent**
- b) $P(C \text{---} A) = P(C \text{---} \neg A) = 0.3$ means C doesn't depend on A's value. Therefore, A and C **ARE independent**.

Problem 5

- a) $P(+o) = 0.24 + 0.06 + 0.08 + 0.12 = \mathbf{0.50}$
b) $P(\text{high energy} \text{ --- } +o) = [P(+o, \text{high energy})] / P(+o)$
- $P(+o, \text{high energy}) = 0.24 + 0.08 = 0.32$

- $P(\text{high energy} \text{ — } +o) = 0.32 / 0.50 = \mathbf{0.64}$
- c) $P(\text{high temp} \text{ — } -o, \text{ low energy}) = P(-o, \text{ high temp, low energy}) / P(-o, \text{ low energy})$
- $P(-o, \text{ high temp, low energy}) = 0.10$
- $P(-o, \text{ low energy}) = 0.10 + 0.25 = 0.35$
- $P(\text{high temp} \text{ — } -o, \text{ low energy}) = 0.10 / 0.35 = \mathbf{2/7 \approx 0.286}$

Problem 6

$P(\text{Wet})$ = sum over all combinations of Rain and Sprinkler:

$$P(Wet) = P(W|R, S) \cdot P(S|R) \cdot P(R) \quad (6)$$

$$+ P(W|R, \neg S) \cdot P(\neg S|R) \cdot P(R) \quad (7)$$

$$+ P(W|\neg R, S) \cdot P(S|\neg R) \cdot P(\neg R) \quad (8)$$

$$+ P(W|\neg R, \neg S) \cdot P(\neg S|\neg R) \cdot P(\neg R) \quad (9)$$

Calculating:

- $0.99 \times 0.01 \times 0.2 = 0.00198$
- $0.80 \times 0.99 \times 0.2 = 0.1584$
- $0.90 \times 0.4 \times 0.8 = 0.288$
- $0.01 \times 0.6 \times 0.8 = 0.0048$

$$P(Wet) = 0.00198 + 0.1584 + 0.288 + 0.0048 = \mathbf{0.45318}$$

Problem 7

$P(D)$ requires marginalizing over A, B, and C:

For each combination:

- $P(A, B, C, D): 0.3 \times 0.8 \times 0.7 \times 0.9 = 0.1512$
- $P(A, B, \neg C, D): 0.3 \times 0.8 \times 0.3 \times 0.6 = 0.0432$
- $P(A, \neg B, C, D): 0.3 \times 0.2 \times 0.7 \times 0.5 = 0.021$
- $P(A, \neg B, \neg C, D): 0.3 \times 0.2 \times 0.3 \times 0.1 = 0.0018$
- $P(\neg A, B, C, D): 0.7 \times 0.2 \times 0.4 \times 0.9 = 0.0504$
- $P(\neg A, B, \neg C, D): 0.7 \times 0.2 \times 0.6 \times 0.6 = 0.0504$
- $P(\neg A, \neg B, C, D): 0.7 \times 0.8 \times 0.4 \times 0.5 = 0.112$
- $P(\neg A, \neg B, \neg C, D): 0.7 \times 0.8 \times 0.6 \times 0.1 = 0.0336$

$$P(D) = 0.1512 + 0.0432 + 0.021 + 0.0018 + 0.0504 + 0.0504 + 0.112 + 0.0336 = \mathbf{0.4636}$$

Problem 8

a) Definition: $P(C \text{---} R) = P(C, R) / P(R)$

- $P(\text{yes---yes}) = 0.32 / 0.50 = \mathbf{0.64}$
- $P(\text{no---yes}) = 0.18 / 0.50 = \mathbf{0.36}$

b) Normalization trick:

- Select rows with $R=\text{yes}$: (yes,yes):0.32, (no,yes):0.18
- Sum = 0.50
- Normalize: $0.32/0.50 = \mathbf{0.64}$, $0.18/0.50 = \mathbf{0.36}$

Problem 9

$$P(A, B, C) = P(C \text{---} A, B) \times P(B \text{---} A) \times P(A) = 0.8 \times 0.7 \times 0.4 = \mathbf{0.224}$$

Problem 10

a) $P(\text{Won't Start})$:

$$= 1 \times P(B, G) + 1 \times P(B, \neg G) + 1 \times P(\neg B, G) + 0.05 \times P(\neg B, \neg G) \quad (10)$$

$$= P(B)P(G) + P(B)P(\neg G) + P(\neg B)P(G) + 0.05 \times P(\neg B)P(\neg G) \quad (11)$$

$$= 0.3 \times 0.1 + 0.3 \times 0.9 + 0.7 \times 0.1 + 0.05 \times 0.7 \times 0.9 \quad (12)$$

$$= 0.03 + 0.27 + 0.07 + 0.0315 = \mathbf{0.4015} \quad (13)$$

b) $P(B \text{---} \text{Won't Start}) = [P(\text{Won't Start---} B) \times P(B)] / P(\text{Won't Start}) = [1 \times 0.3] / 0.4015 = \mathbf{0.747}$

c) $P(B \text{---} \text{Won't Start}, \neg G) = P(\text{Won't Start---} B, \neg G) \times P(B) / P(\text{Won't Start}, \neg G) = [1 \times 0.3] / [0.27 + 0.0315] = 0.3 / 0.3015 = \mathbf{0.995}$